

PAPERMIND CROSS-PAGE DEMO EXAM

Section A: Operating Systems

1. Which component of an OS is loaded into memory first during boot?

- (A) Shell
- (B) Kernel
- (C) Desktop GUI
- (D) Web Browser

2. In modern multi-threaded operating systems, describe how synchronization is achieved between con

- (A) Mutex locks and counting semaphores
- (B) Infinite spin loops with no yield
- (C) Random process termination
- (D) Disabling all hardware interrupts permanently

3. Which scheduling algorithm may lead to starvation of low-priority tasks?

- (A) Round Robin
- (B) First-Come First-Served
- (C) Strict Priority Scheduling
- (D) Fair Share Scheduling